LTspice XVII - OTA_TANMAY_final

2 Stage OTA Simulations

Roll no. 240102111

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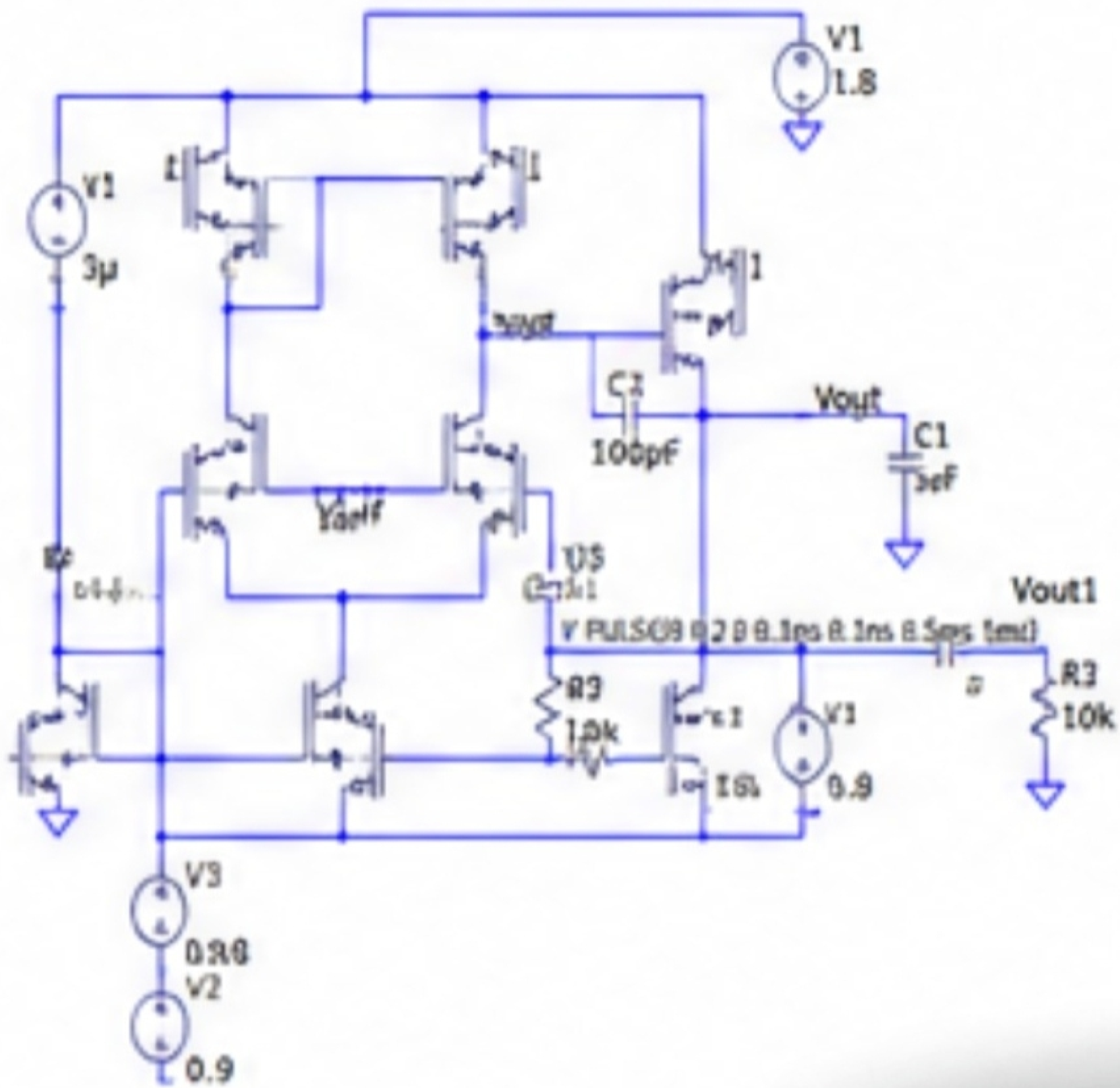
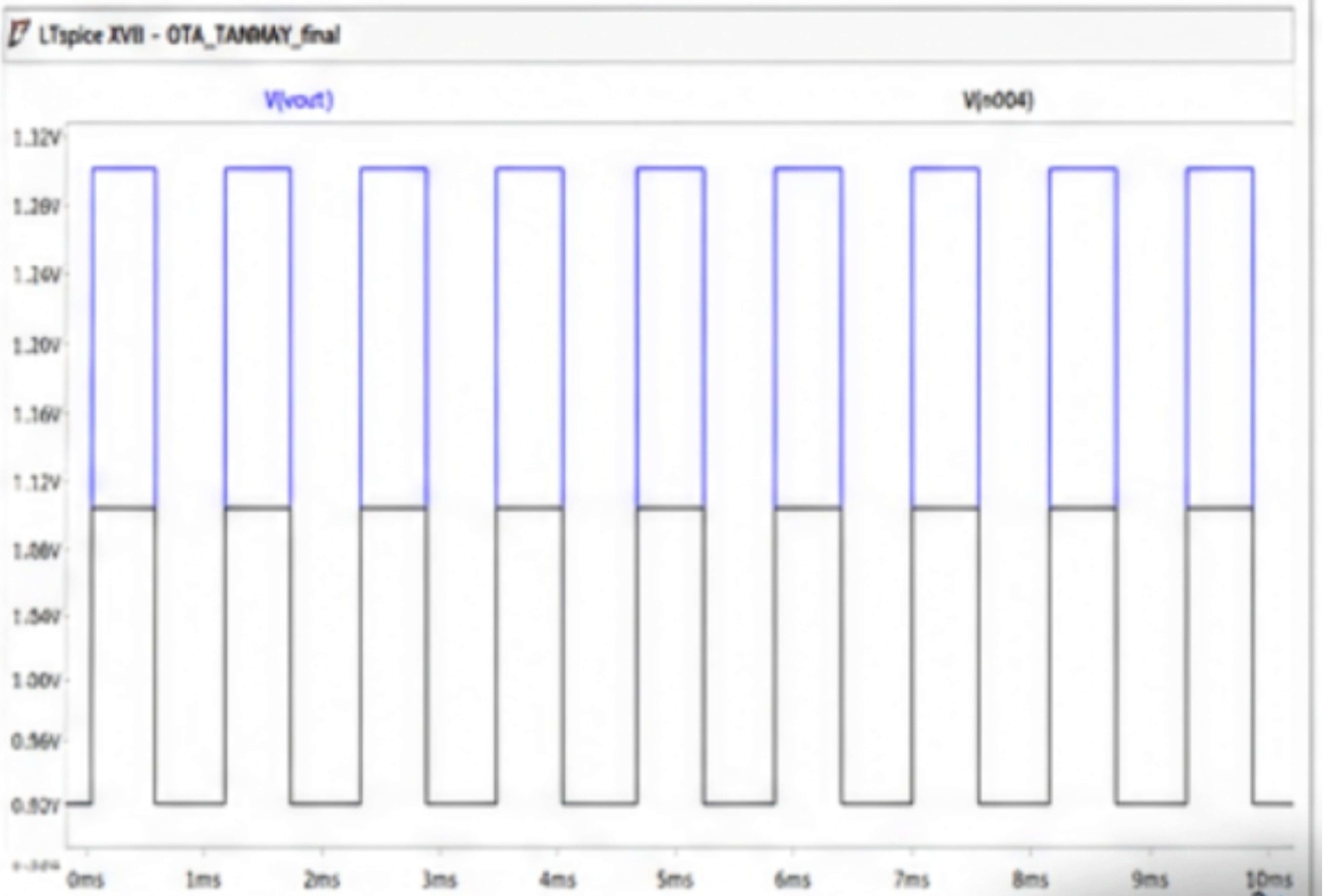


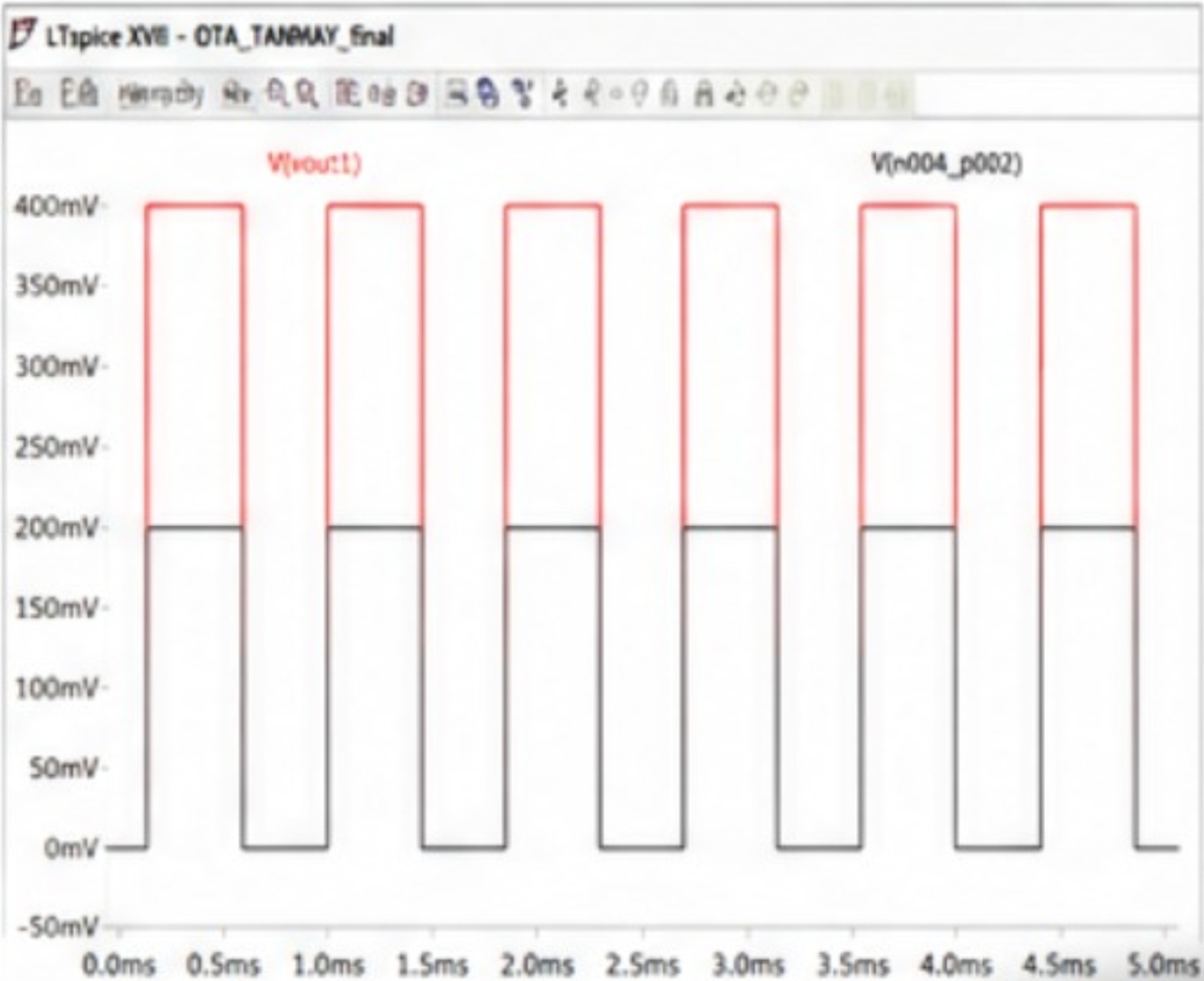
Non inverting amplifier with gain 2

.tran 5ms

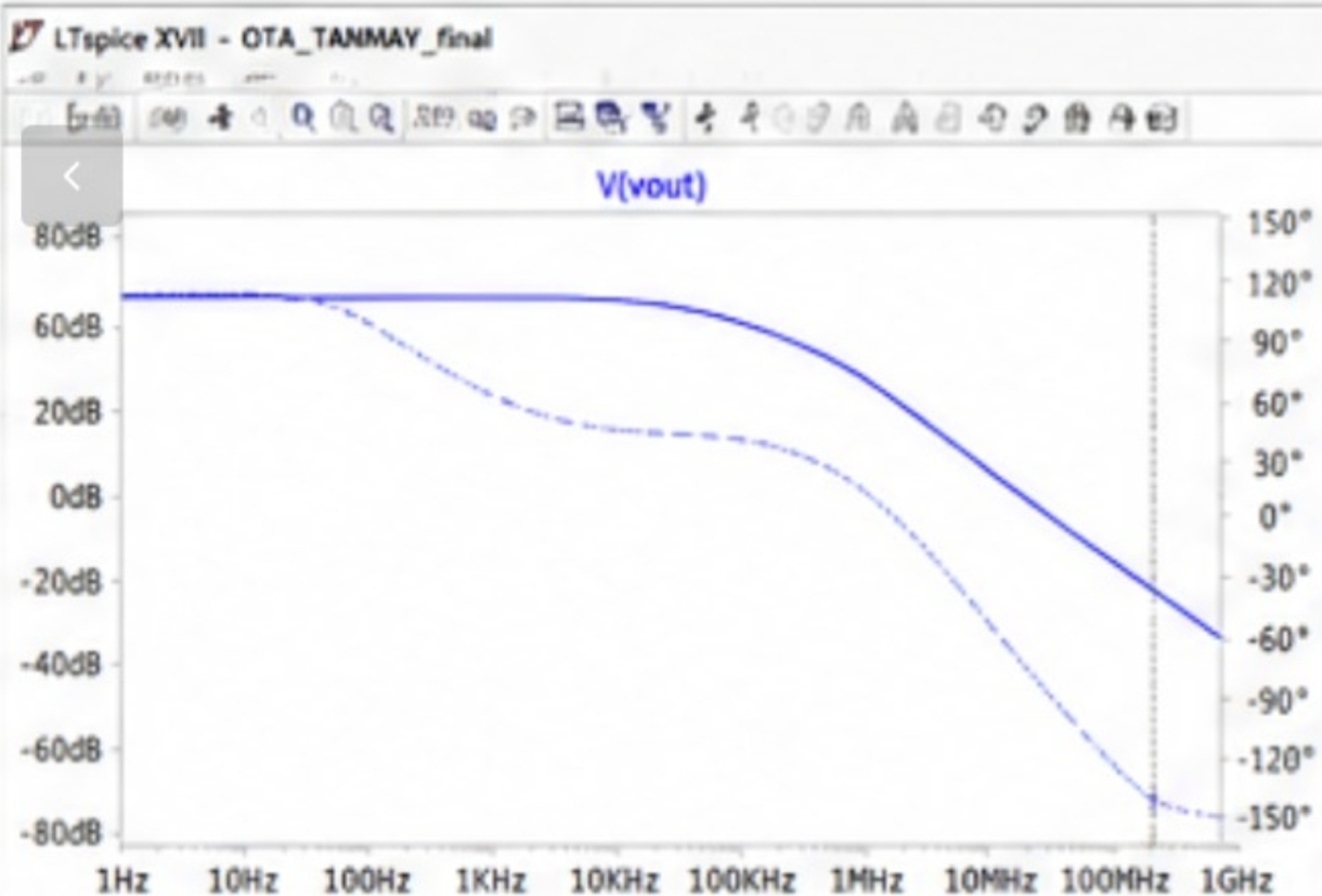
Simulation with Input pulse of 0.2V(black) giving output 0.4V(Blue)......at terminal Vout



Output taken at terminal Vout1



DC Gain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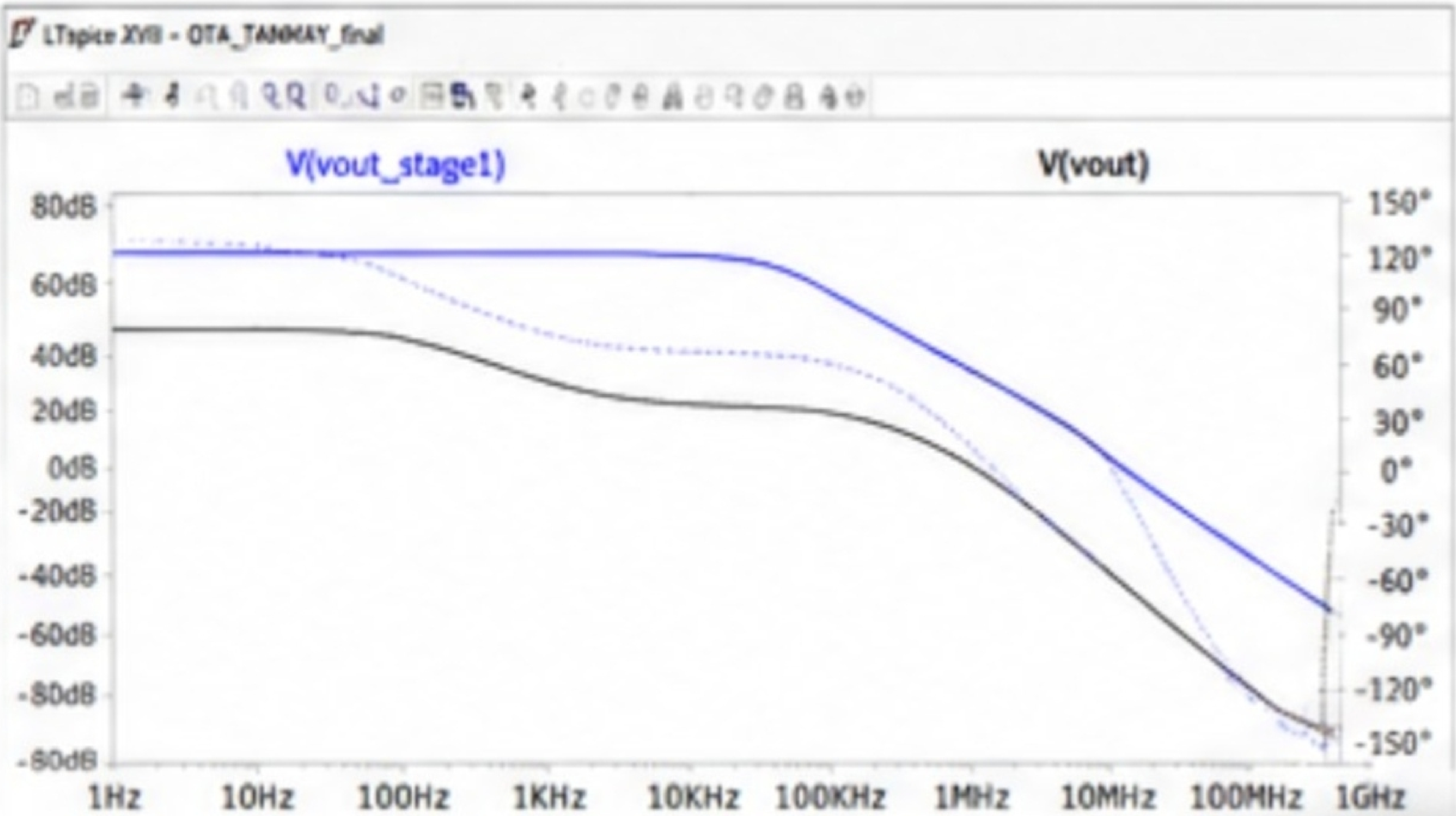
V(vout)
80dB
60dB
20dB
0dB
-20dB
-40dB
-60dB
-80dB
150°
120°
90°
60°
30°
0°
-30°
-60°
-90°
-120°
-150°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz
100MHz
1GHz

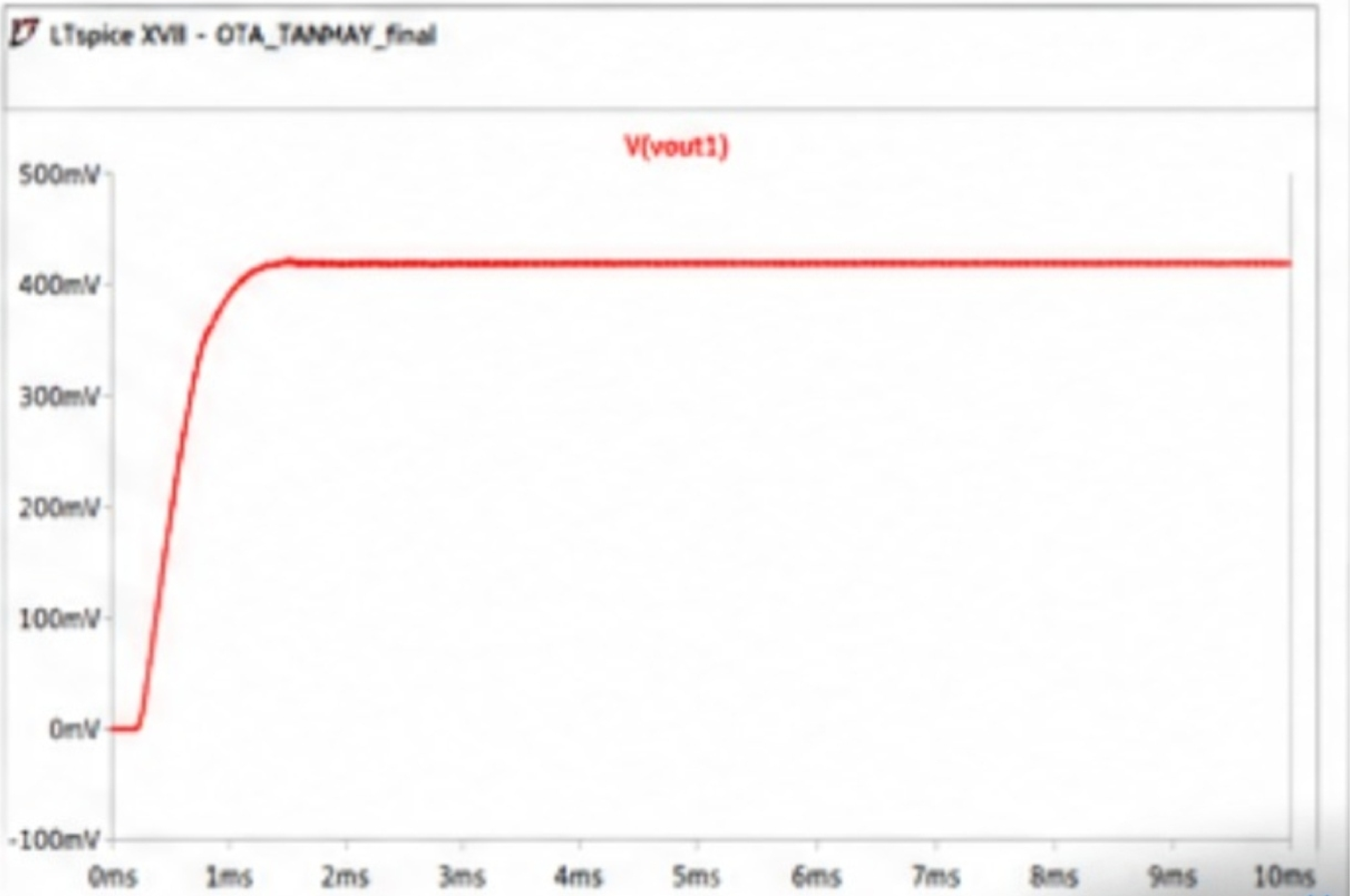
DC Gain

Blue : Stage 1 DC Gain

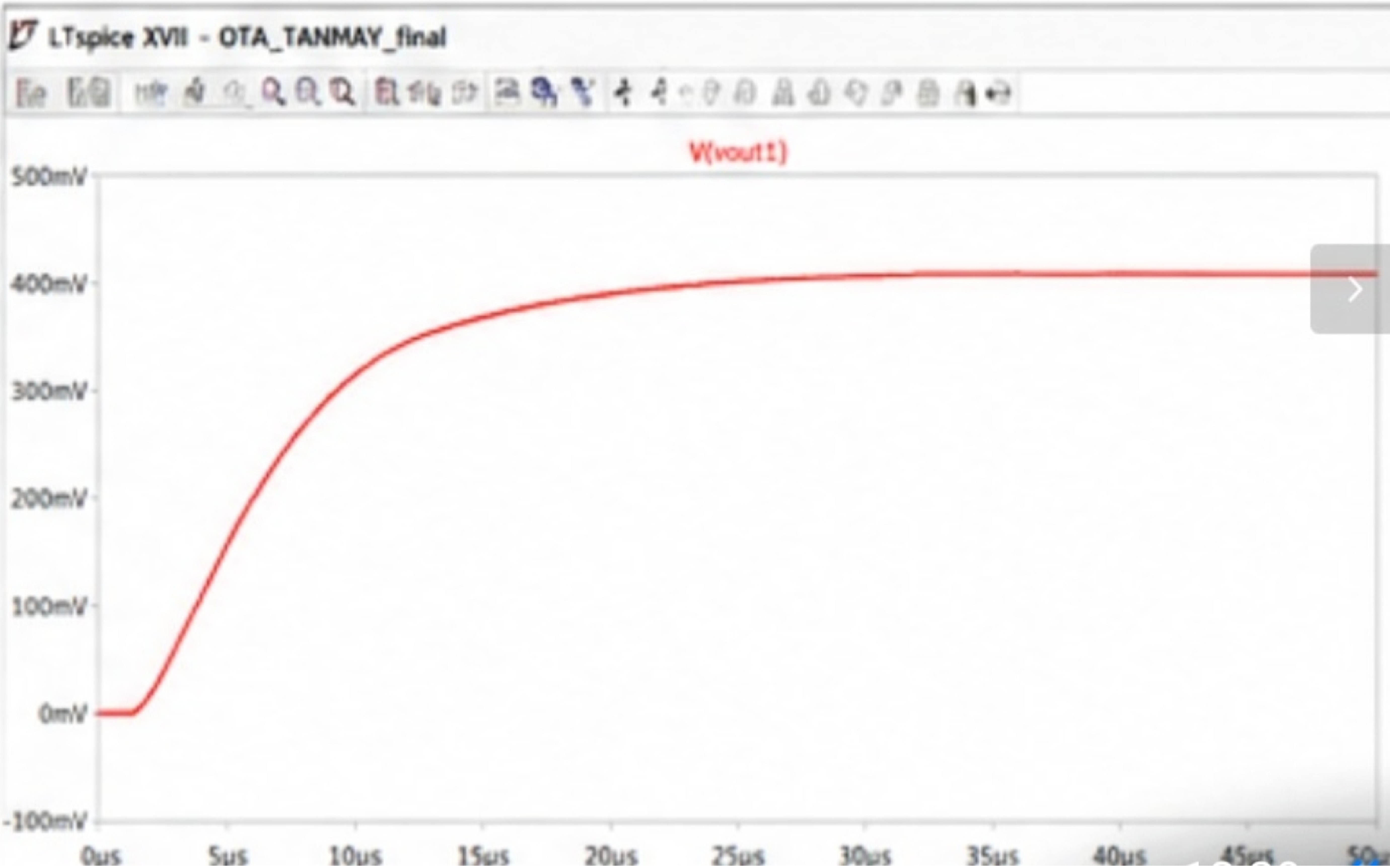
Black : Net DC Gain



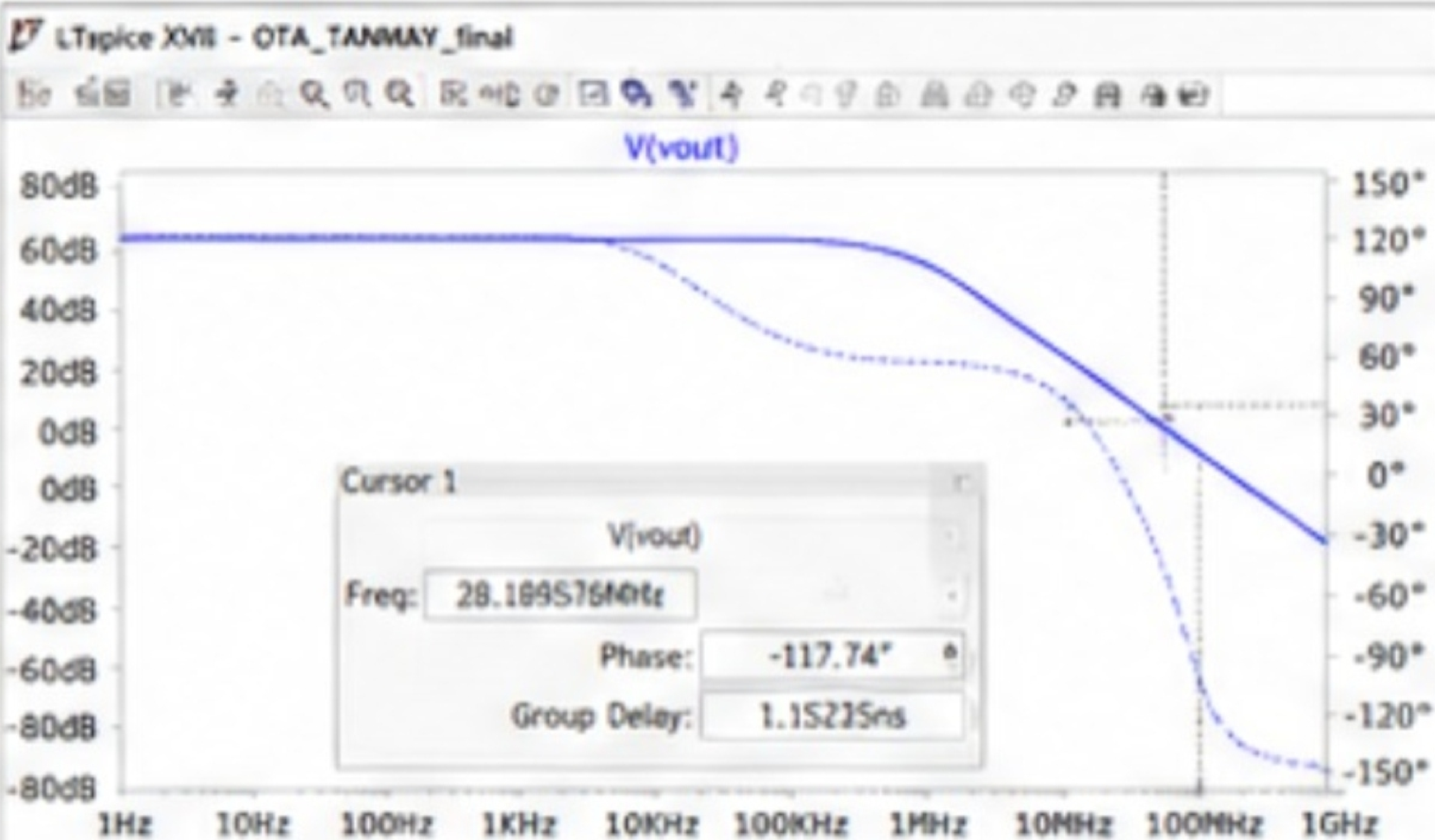
Transient Response of Closed loop amplifier with step input 0.2V



Transient Response of Closed loop amplifier with step input 0.2V



Phase Margin



Phase margin :

180 – 117.74 = 62.26

DC operating points

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--- Operating Point ---

V(vdd):	1.80000	voltage
V(vdiff):	899.99787m	voltage
V(vout):	400.00021m	voltage
V(vout1):	399.99958m	voltage
V(n004):	1.0999998	voltage
V(n004_p002):	200.00002m	voltage
Id(M0):	1.500000µ	device_current
Id(M1):	1.500000µ	device_current
Id(M2):	750.00009n	device_current
Id(M3):	750.00009n	device_current
Id(M4):	1.500000µ	device_current
Id(M5):	750.00009n	device_current
Id(M6):	750.00009n	device_current
Id(M7):	1.500000µ	device_current

DC operating points

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I(I1):	3.000000µ	device_current	V2@branch:	90.000000µ	current
I(V1):	-3.000000µ	device_current	V3@branch:	-1.500000µ	current
I(V2):	90.000000µ	device_current	I(C1):	-22.611500p	device_current
I(V3):	-1.500000µ	device_current	I(C2):	-570.45126p	device_current
			I(C3):	-2.2611508n	device_current